A Monero data model and API specification

implemented by monero-kotlin, for Kotlin
using monero v0.18.5.4 'Fluorine Fermi'

Last updated November 6, 2026



Kotlin implementation: https://github.com/libmonero/monero-kotlin

C ABI used by the wallet and the daemon: https://github.com/libmonero/monero-c
C# implementation: https://github.com/libmonero/monero-csharp
Python implementation: https://github.com/libmonero/monero-python
Java implementation: https://github.com/woodser/monero-java
TypeScript implementation: https://github.com/woodser/monero-ts
C++ implementation: https://github.com/woodser/monero-cpp

Document source: https://github.com/libmonero/monero-kotlin/blob/main/docs/monero-spec.drawio

Document created and revisable using: https://www.draw.io/

How to read the Kotlin tables
The rows are the public members of MoneroDaemon and MoneroWallet, and of the model classes, in Kotlin types. The calls are blocking: a function that talks to the node or to the wallet returns its result, or throws a MoneroError. A property with a ? is nullable, and null means that the value is not known. Amounts are in atomic units, as BigInteger.

Monero Daemon Interface 1/3 - MoneroDaemon

addListener(listener: MoneroDaemonListener)

removeListener(listener: MoneroDaemonListener)

getListeners(): List<MoneroDaemonListener>

removeListeners()

getVersion(): MoneroVersion

isTrusted(): Boolean

getHeight(): Long

getInfo(): MoneroDaemonInfo

getSyncInfo(): MoneroDaemonSyncInfo

getNetworkStats(): MoneroDaemonNetworkStats

getHardForkInfo(): MoneroHardForkInfo

getAltChains(): List<MoneroAltChain>

getAltBlockHashes(): List<String>

waitForNextBlockHeader(): MoneroBlockHeader

getBlockHash(height: Long): String

getLastBlockHeader(): MoneroBlockHeader

getBlockHeaderByHash(blockHash: String): MoneroBlockHeader

getBlockHeaderByHeight(height: Long): MoneroBlockHeader

getBlockHeadersByRange(startHeight: Long, endHeight: Long): List<MoneroBlockHeader>

getBlockByHash(blockHash: String): MoneroBlock

getBlockByHeight(height: Long): MoneroBlock

getBlocksByHash(blockHashes: List<String>, startHeight: Long, prune: Boolean = false,
    maxBlockCount: Long = 0): MoneroGetBlocksByHashResult

getBlocksByHeight(heights: List<Long>): List<MoneroBlock>

getBlocksByRange(startHeight: Long? = null, endHeight: Long? = null): List<MoneroBlock>

getBlocksByRangeChunked(startHeight: Long? = null, endHeight: Long? = null,
    maxChunkSize: Long? = null): List<MoneroBlock>

getBlockHashes(blockHashes: List<String>): MoneroGetBlockHashesResult

getTx(txHash: String, prune: Boolean = false): MoneroTx?

getTxs(txHashes: Collection<String>, prune: Boolean = false): List<MoneroTx>

getTxHex(txHash: String, prune: Boolean = false): String

getTxHexes(txHashes: Collection<String>, prune: Boolean = false): List<String>

submitTxHex(txHex: String, doNotRelay: Boolean = false): MoneroSubmitTxResult

relayTxByHash(txHash: String)

relayTxsByHash(txHashes: Collection<String>)

Monero Daemon Interface 2/3 - MoneroDaemon

```
getTxPool(): List<MoneroTx>

getTxPoolHashes(): List<String>

getTxPoolBacklog(): List<MoneroTxBacklogEntry>

getTxPoolStats(): MoneroTxPoolStats

flushTxPool()

flushTxPool(hash: String)

flushTxPool(hashes: Collection<String>)

getKeyImageSpentStatus(keyImage: String): MoneroKeyImageSpentStatus

getKeyImageSpentStatuses(keyImages: List<String>): List<MoneroKeyImageSpentStatus>

getOutputIndices(txHash: String): List<Long>

getOutputs(outputs: Collection<MoneroOutput>): List<MoneroOutput>

getOutputHistogram(amounts: Collection<BigInteger> = emptyList(), minCount: Int? = null, maxCount: Int? = null,
      isUnlocked: Boolean? = null, recentCutoff: Int? = null): List<MoneroOutputHistogramEntry>

getOutputDistribution(amounts: Collection<BigInteger>, isCumulative: Boolean? = null, startHeight: Long? = null,
      endHeight: Long? = null): List<MoneroOutputDistributionEntry>

getFeeEstimate(graceBlocks: Long = 0): MoneroFeeEstimate

getBlockTemplate(walletAddress: String, reserveSize: Int? = null): MoneroBlockTemplate

getMinerData(): MoneroMinerData

calculatePow(majorVersion: Int, height: Long, blockBlob: String, seedHash: String): String

addAuxiliaryPow(blockTemplateBlob: String, auxPow: List<MoneroAuxiliaryPow>): MoneroAddAuxiliaryPowResult

getMinerTxSum(height: Long, numBlocks: Long): MoneroMinerTxSum

startMining(address: String, numThreads: Long? = null, isBackground: Boolean? = null, ignoreBattery: Boolean? = null)

stopMining()

getMiningStatus(): MoneroMiningStatus

generateBlocks(address: String, numBlocks: Long, prevBlockHash: String? = null,
      startingNonce: Long? = null): MoneroGenerateBlocksResult

submitBlock(blockBlob: String)

submitBlocks(blockBlobs: Collection<String>)

getPeers(): List<MoneroPeer>

getKnownPeers(): List<MoneroPeer>

getPublicPeers(includeOffline: Boolean = false): List<MoneroPeer>

setOutgoingPeerLimit(limit: Int)

setIncomingPeerLimit(limit: Int)

getPeerBans(): List<MoneroBan>

getPeerBan(address: String): MoneroBan

setPeerBans(bans: List<MoneroBan>)

setPeerBan(ban: MoneroBan)

getDownloadLimit(): Int
```

Monero Daemon Interface 3/3 - MoneroDaemon

setDownloadLimit(limit: Int): Int

resetDownloadLimit(): Int

getUploadLimit(): Int

setUploadLimit(limit: Int): Int

resetUploadLimit(): Int

pruneBlockchain(check: Boolean = false): MoneroPruneResult

saveBlockchain()

popBlocks(numBlocks: Long): Long

flushCache(badBlocks: Boolean = false)

setBootstrapDaemon(address: String, username: String = "", password: String = "", proxy: String = "")

removeBootstrapDaemon()

setLogHashRate(isVisible: Boolean)

setLogLevel(level: Int)

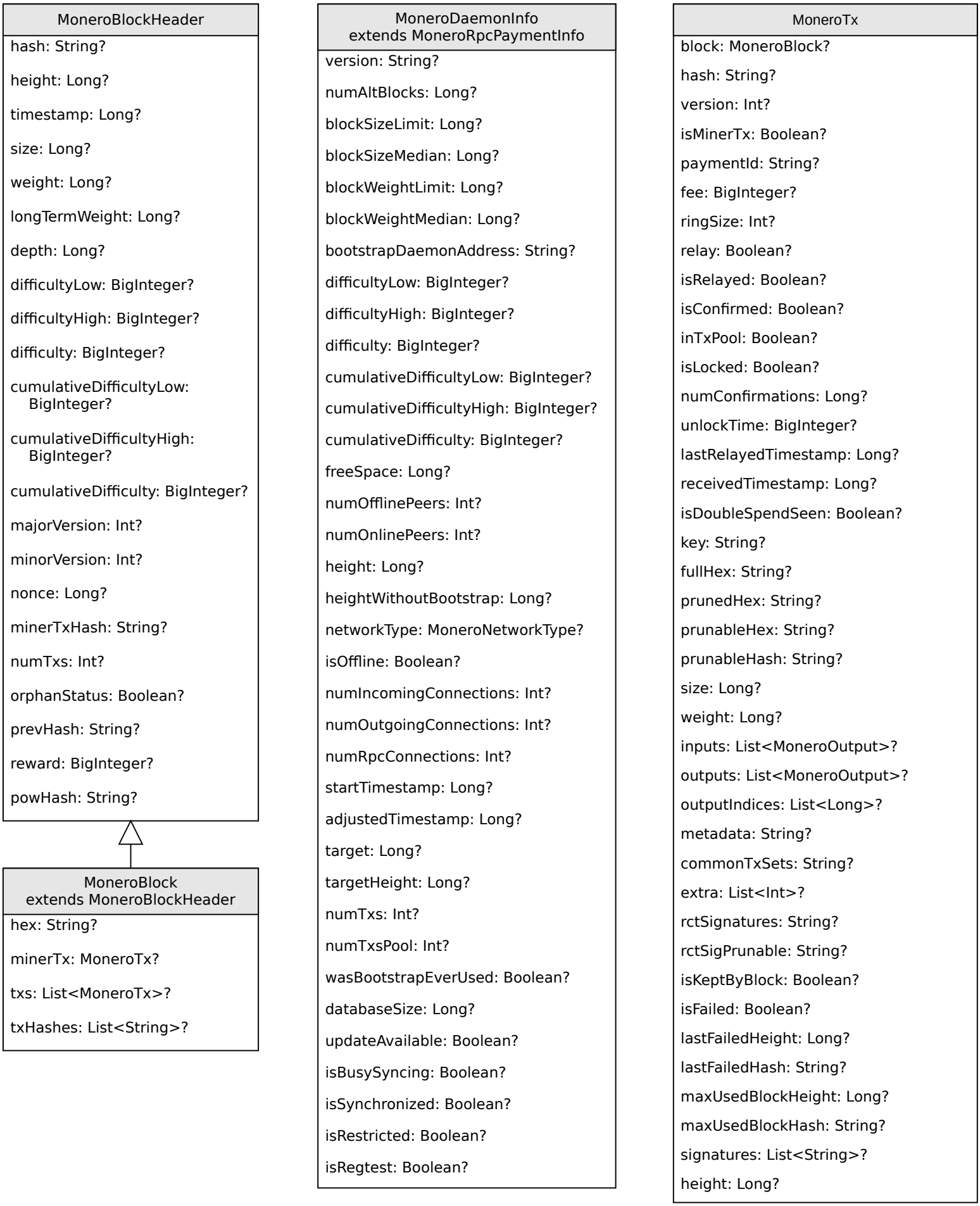
setLogCategories(categories: String = ""): String

checkForUpdate(): MoneroDaemonUpdateCheckResult

downloadUpdate(path: String = ""): MoneroDaemonUpdateDownloadResult

stop()

Monero Daemon Types 1/3



Monero Daemon Types 1/3 (continued)

MoneroOutput
tx: MoneroTx?
keyImage: MoneroKeyImage?
amount: BigInteger?
index: Long?
ringOutputIndices: List<Long>?
stealthPublicKey: String?
mask: String?

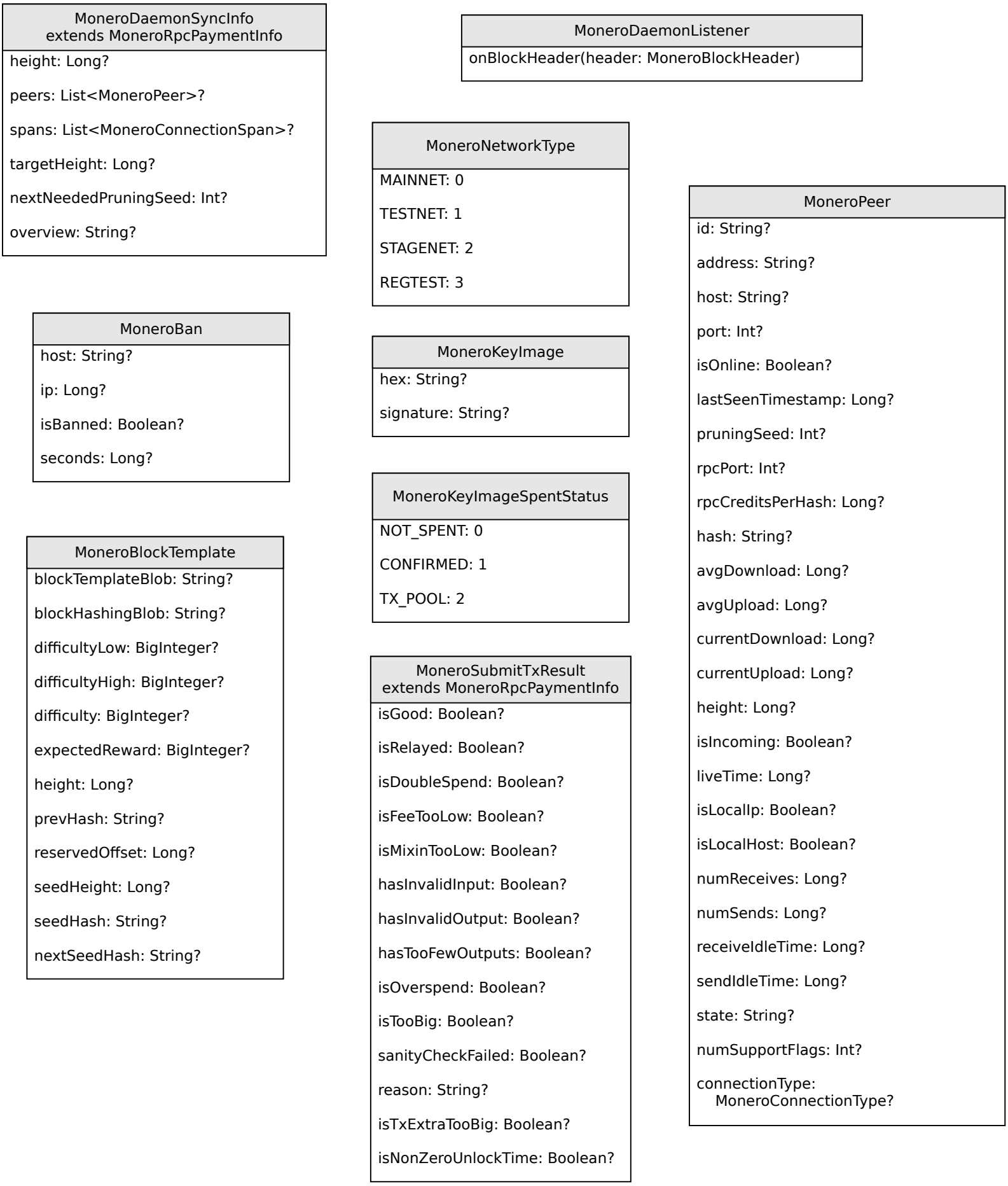
MoneroBlocksResult
blocks: List<MoneroBlock>?

MoneroGetBlockHashesResult
startHeight: Long?
currentHeight: Long?
hashes: List<String>?

MoneroGetBlocksByHashResult
currentHeight: Long?
blocks: List<MoneroBlock>?

MoneroRpcPaymentInfo
credits: Long?
topBlockHash: String?

Monero Daemon Types 2/3



Monero Daemon Types 2/3 (continued)

MoneroConnectionSpan
connectionId: String?
numBlocks: Long?
remoteAddress: String?
rate: Long?
speed: Long?
size: Long?
startHeight: Long?

MoneroAddAuxiliaryPowResult extends MoneroRpcPaymentInfo
blockTemplateBlob: String?
blockHashingBlob: String?
merkleRoot: String?
merkleTreeDepth: Int?
auxPow: List<MoneroAuxiliaryPow>?

MoneroAuxiliaryPow
id: String?
hash: String?

MoneroGenerateBlocksResult
height: Long?
blockHashes: List<String>?

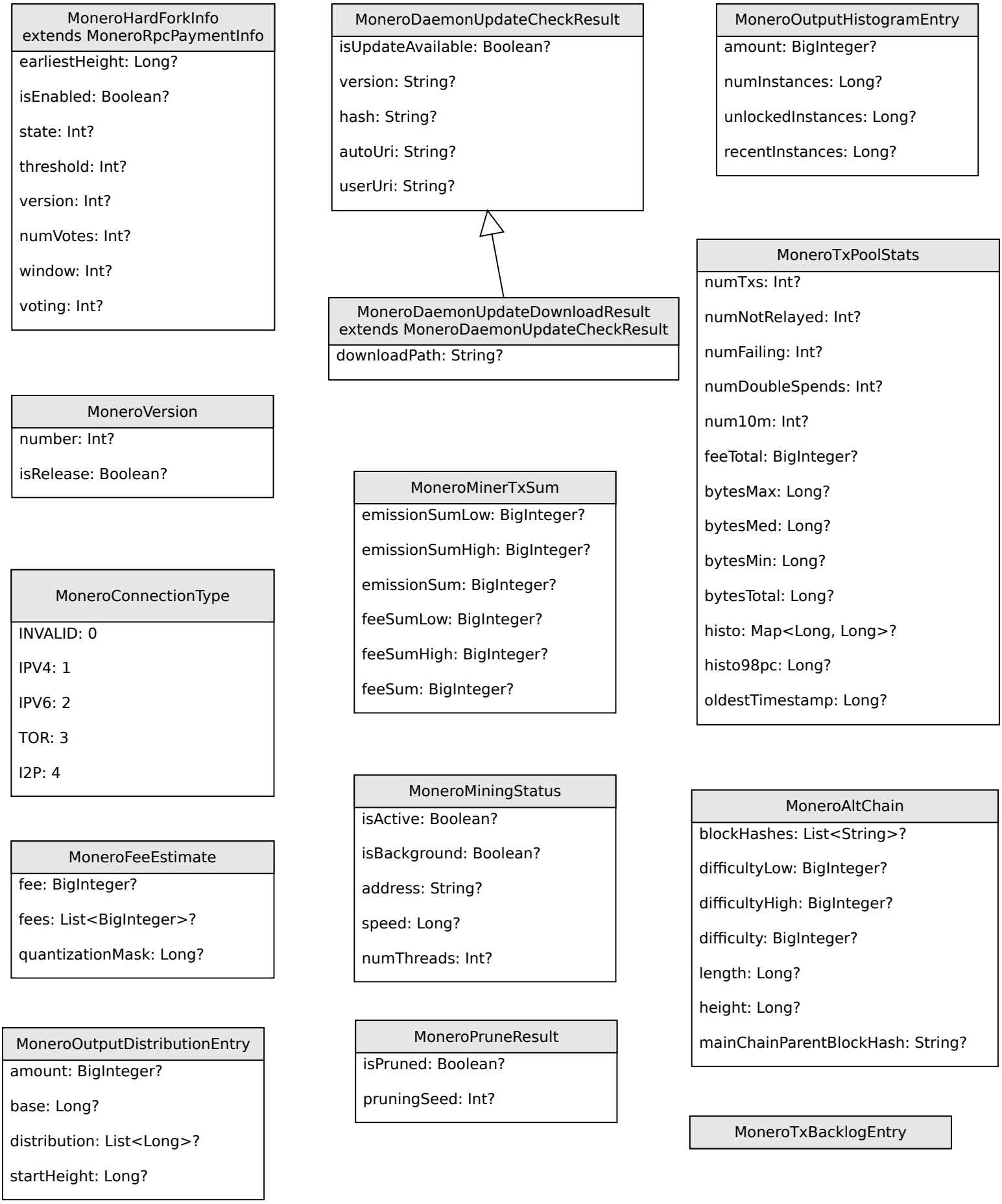
MoneroMinerData extends MoneroRpcPaymentInfo
majorVersion: Int?
height: Long?
prevHash: String?
seedHash: String?
difficulty: String?
medianWeight: Long?
alreadyGeneratedCoins: BigInteger?
txPoolBacklog: List<MoneroTx>?

Monero Daemon Types 2/3 (continued)

MoneroRpcConnection
uri: String?
username: String?
password: String?
proxyUri: String?
zmqUri: String?
priority: Int?
timeoutMs: Long?
responseTime: Long?
sslVerify: Boolean?
isOnline: Boolean?
isAuthenticated: Boolean?

MoneroDaemonNetworkStats extends MoneroRpcPaymentInfo
startTime: Long?
totalPacketsIn: Long?
totalBytesIn: Long?
totalPacketsOut: Long?
totalBytesOut: Long?

Monero Daemon Types 3/3



Monero Wallet Interface 1/4 - MoneroWallet

addListener(listener: MoneroWalletListener)

removeListener(listener: MoneroWalletListener)

getListeners(): List<MoneroWalletListener>

removeListeners()

getVersion(): MoneroVersion

getPath(): String

getNetworkType(): MoneroNetworkType

isViewOnly(): Boolean

isClosed(): Boolean

getSeed(): String

getSeedLanguage(): String

getPrivateViewKey(): String

getPrivateSpendKey(): String

getPublicViewKey(): String

getPublicSpendKey(): String

getPrimaryAddress(): String

getAddress(accountIdx: Int, subaddressIdx: Int): String

getAddressIndex(address: String): MoneroSubaddress

getIntegratedAddress(standardAddress: String? = null, paymentId: String? = null): MoneroIntegratedAddress

decodeIntegratedAddress(integratedAddress: String): MoneroIntegratedAddress

getAttribute(key: String): String?

setAttribute(key: String, value: String)

getDefaultFeePriority(): MoneroTxPriority

save()

moveTo(path: String, password: String)

changePassword(oldPassword: String, newPassword: String)

close(save: Boolean)

getDaemonConnection(): MoneroRpcConnection?

setDaemonConnection(connection: MoneroRpcConnection?, isTrusted: Boolean? = null)

setDaemonConnection(uri: String, username: String? = null, password: String? = null, proxyUri: String? = null,
isTrusted: Boolean? = null)

isConnectedToDaemon(): Boolean

isDaemonSynced(): Boolean

isDaemonTrusted(): Boolean

isSynced(): Boolean

getHeight(): Long

getDaemonHeight(): Long

Monero Wallet Interface 2/4 - MoneroWallet

getDaemonMaxPeerHeight(): Long

getHeightByDate(year: Int, month: Int, day: Int): Long

getRestoreHeight(): Long

setRestoreHeight(restoreHeight: Long)

sync(startHeight: Long? = null, listener: MoneroWalletListener? = null): MoneroSyncResult

sync(listener: MoneroWalletListener): MoneroSyncResult

startSyncing(syncPeriodInMs: Long = 10000)

stopSyncing()

waitForNextBlock(): Long

scanTxs(txHashes: List<String>)

rescanSpent()

rescanBlockchain()

startMining(numThreads: Long? = null, backgroundMining: Boolean? = null, ignoreBattery: Boolean? = null)

stopMining()

getBalance(accountIdx: Int? = null, subaddressIdx: Int? = null): BigInteger

getUnlockedBalance(accountIdx: Int? = null, subaddressIdx: Int? = null): BigInteger

getAccounts(includeSubaddresses: Boolean = false, tag: String? = null): List<MoneroAccount>

getAccount(accountIdx: Int, includeSubaddresses: Boolean = false): MoneroAccount

createAccount(label: String? = null): MoneroAccount

getAccountTags(): List<MoneroAccountTag>

tagAccounts(tag: String, accountIndices: List<Int>)

untagAccounts(accountIndices: List<Int>)

setAccountTagLabel(tag: String, label: String)

getSubaddresses(accountIdx: Int, subaddressIndices: List<Int>? = null): List<MoneroSubaddress>

getSubaddress(accountIdx: Int, subaddressIdx: Int): MoneroSubaddress

createSubaddress(accountIdx: Int, label: String? = null): MoneroSubaddress

setSubaddressLabel(accountIdx: Int, subaddressIdx: Int, label: String? = null)

getTxs(query: MoneroTxQuery? = null): List<MoneroTxWallet>

getTxs(txHashes: List<String>): List<MoneroTxWallet>

getTx(txHash: String): MoneroTxWallet?

getTransfers(query: MoneroTransferQuery? = null): List<MoneroTransfer>

getTransfers(accountIdx: Int, subaddressIdx: Int? = null): List<MoneroTransfer>

getOutputs(query: MoneroOutputQuery? = null): List<MoneroOutputWallet>

exportOutputs(all: Boolean = false): String

importOutputs(outputsHex: String): Int

exportKeyImages(all: Boolean = false): MoneroKeyImageExportResult

importKeyImages(keyImages: List<MoneroKeyImage>, offset: Long = 0): MoneroKeyImageImportResult

freezeOutput(keyImage: String)

Monero Wallet Interface 3/4 - MoneroWallet

```
thawOutput(keyImage: String)

isOutputFrozen(keyImage: String): Boolean

createTx(config: MoneroTxConfig): MoneroTxWallet

createTxs(config: MoneroTxConfig): List<MoneroTxWallet>

sweepOutput(config: MoneroTxConfig): MoneroTxWallet

sweepUnlocked(config: MoneroTxConfig): List<MoneroTxWallet>

sweepDust(relay: Boolean = false): List<MoneroTxWallet>

relayTx(txMetadata: String): String

relayTx(tx: MoneroTxWallet): String

relayTxs(txMetadatas: List<String>): List<String>

relayTxs(txs: Collection<MoneroTxWallet>): List<String>

describeTxSet(txSet: MoneroTxSet): MoneroTxSet

describeUnsignedTxSet(unsignedTxHex: String): MoneroTxSet

describeMultisigTxSet(multisigTxHex: String): MoneroTxSet

signTxs(unsignedTxHex: String): MoneroTxSet

submitTxs(signedTxHex: String): List<String>

getTxNote(txHash: String): String

getTxNotes(txHashes: List<String>): List<String>

setTxNote(txHash: String, note: String)

setTxNotes(txHashes: List<String>, notes: List<String>)

getAddressBookEntries(indices: List<Long>? = null): List<MoneroAddressBookEntry>

addAddressBookEntry(address: String, description: String? = null): Long

editAddressBookEntry(index: Long, address: String?, description: String?)

deleteAddressBookEntry(index: Long)

getPaymentUri(config: MoneroTxConfig): String

parsePaymentUri(uri: String): MoneroTxConfig

signMessage(message: String,
    signatureType: MoneroMessageSignatureType = MoneroMessageSignatureType.SIGN_WITH_SPEND_KEY,
    accountIdx: Int = 0, subaddressIdx: Int = 0): String

verifyMessage(message: String, address: String, signature: String): MoneroMessageSignatureResult

getTxKey(txHash: String): String

checkTxKey(txHash: String, txKey: String, address: String): MoneroCheckTx

getTxProof(txHash: String, address: String, message: String? = null): String

checkTxProof(txHash: String, address: String, message: String?, signature: String): MoneroCheckTx

getSpendProof(txHash: String, message: String? = null): String

checkSpendProof(txHash: String, message: String?, signature: String): Boolean

getReserveProofWallet(message: String? = null): String

getReserveProofAccount(accountIdx: Int, amount: BigInteger, message: String? = null): String
```

Monero Wallet Interface 4/4 - MoneroWallet

checkReserveProof(address: String, message: String?, signature: String): MoneroCheckReserve

isMultisig(): Boolean

getMultisigInfo(): MoneroMultisigInfo

isMultisigImportNeeded(): Boolean

prepareMultisig(): String

makeMultisig(multisigHexes: List<String>, threshold: Int, password: String): String

exchangeMultisigKeys(multisigHexes: List<String>, password: String): MoneroMultisigInitResult

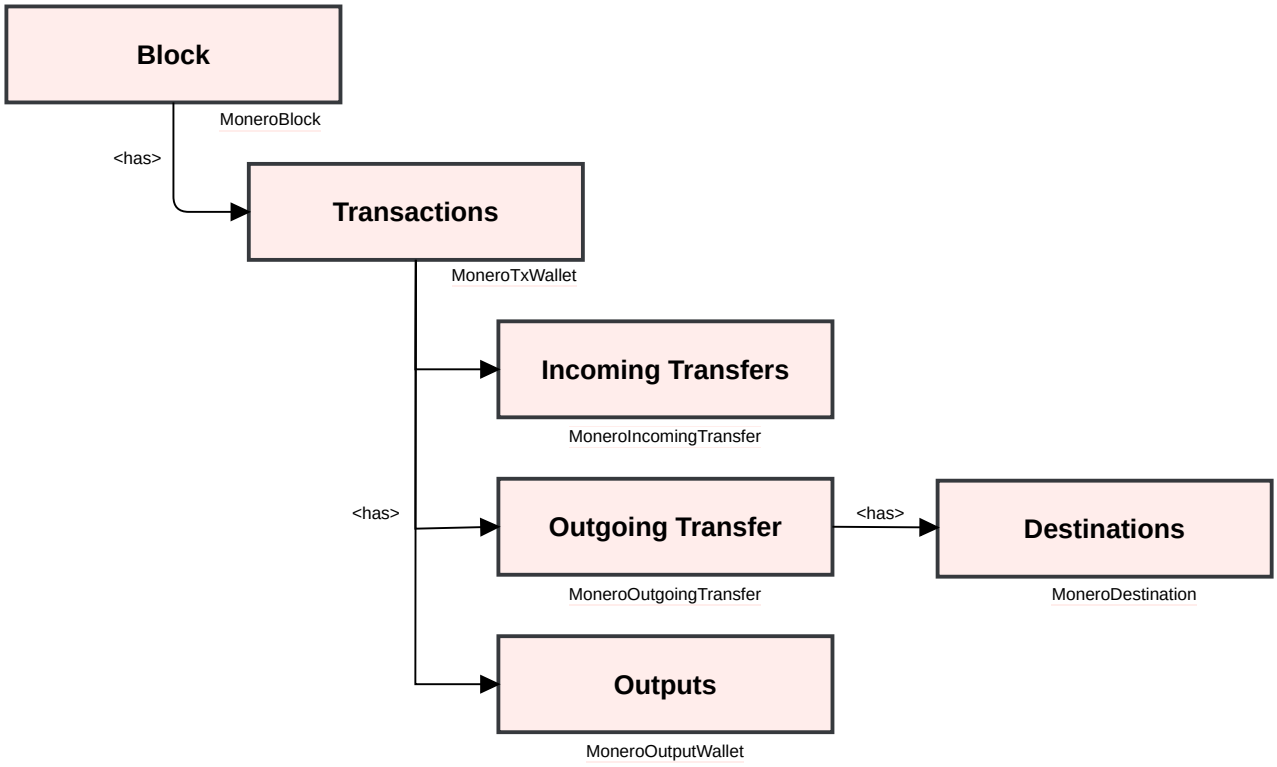
exportMultisigHex(): String

importMultisigHex(multisigHexes: List<String>, refreshAfterImport: Boolean = true): Int

signMultisigTxHex(multisigTxHex: String): MoneroMultisigSignResult

submitMultisigTxHex(signedMultisigTxHex: String): List<String>

Monero Wallet Types 1/4 - Transactions, Transfers, and Outputs



Transactions are part of a block and have incoming transfers and/or an outgoing transfer.

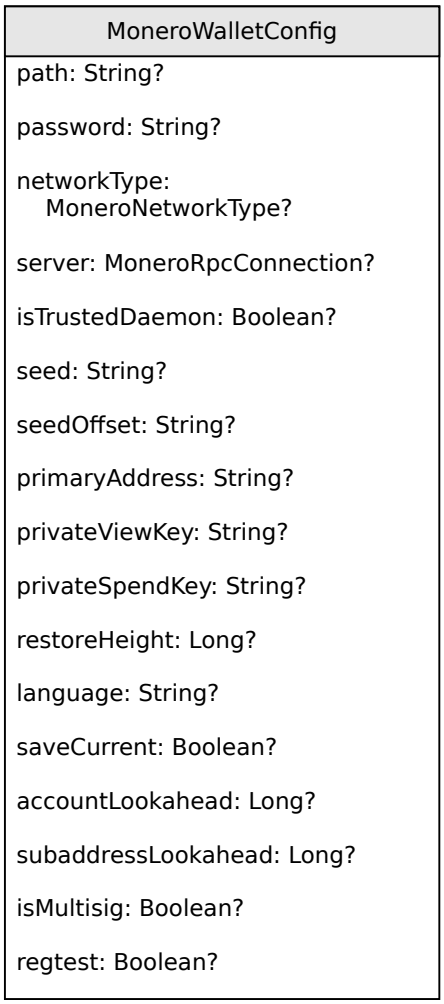
The outgoing transfer may have destination amounts and addresses which are saved locally with the wallet and are not recoverable from the blockchain.

Transactions may also have outputs received by the wallet when requested.

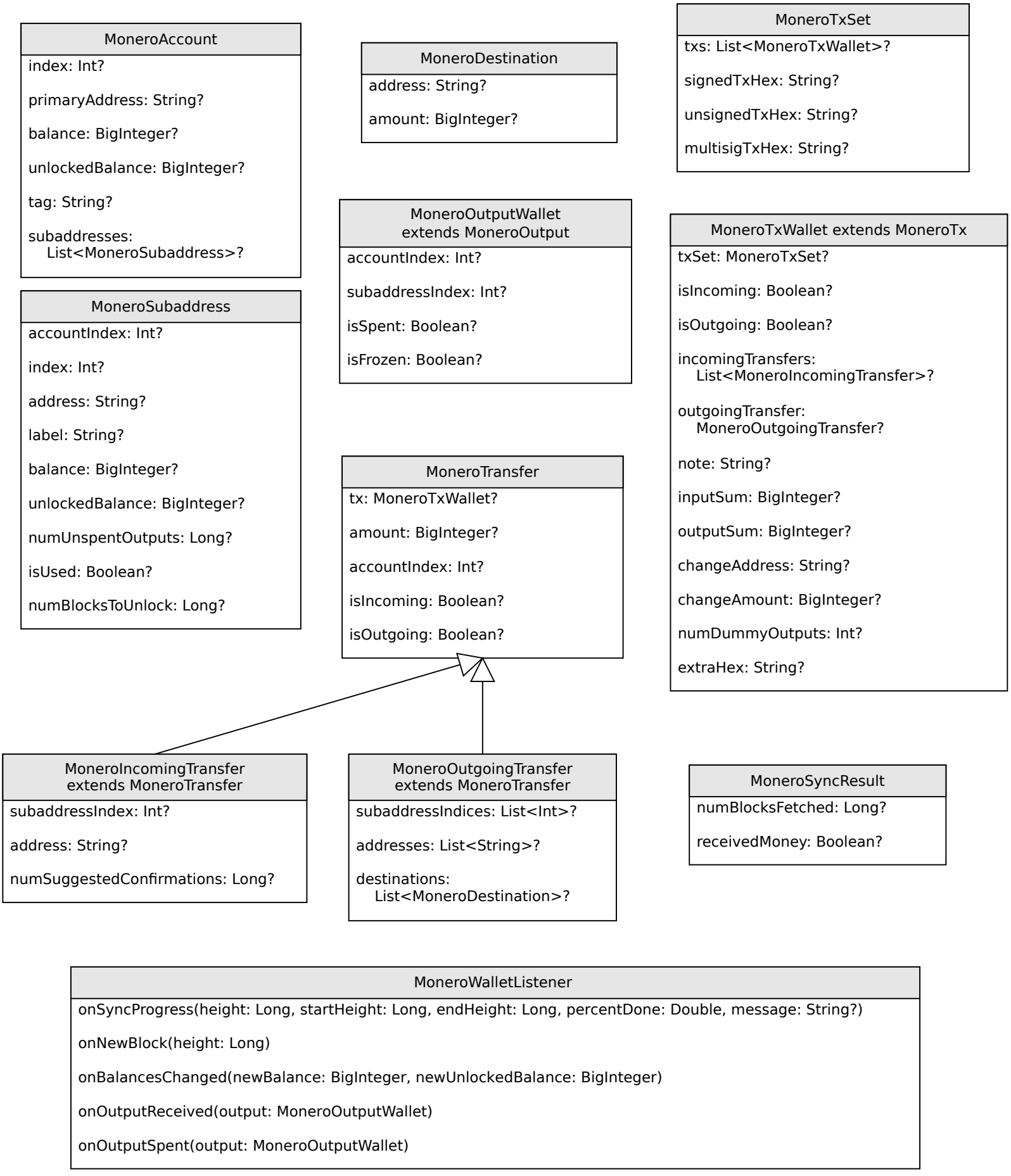
Monero Wallet Types 2/4 - Send Request and Queries for Transactions, Transfers, and Outputs



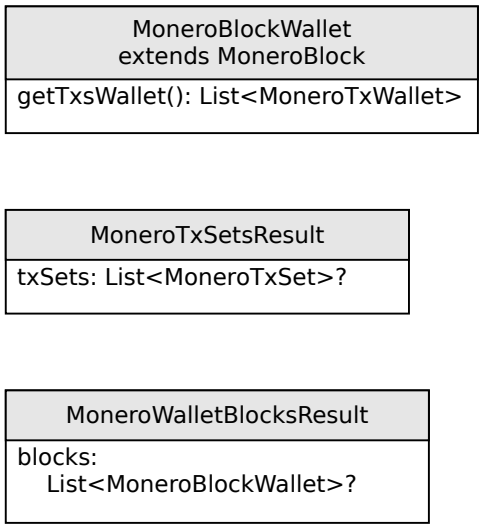
Monero Wallet Types 2/4 - Send Request and Queries for Transactions, Transfers, and Outputs (continued)



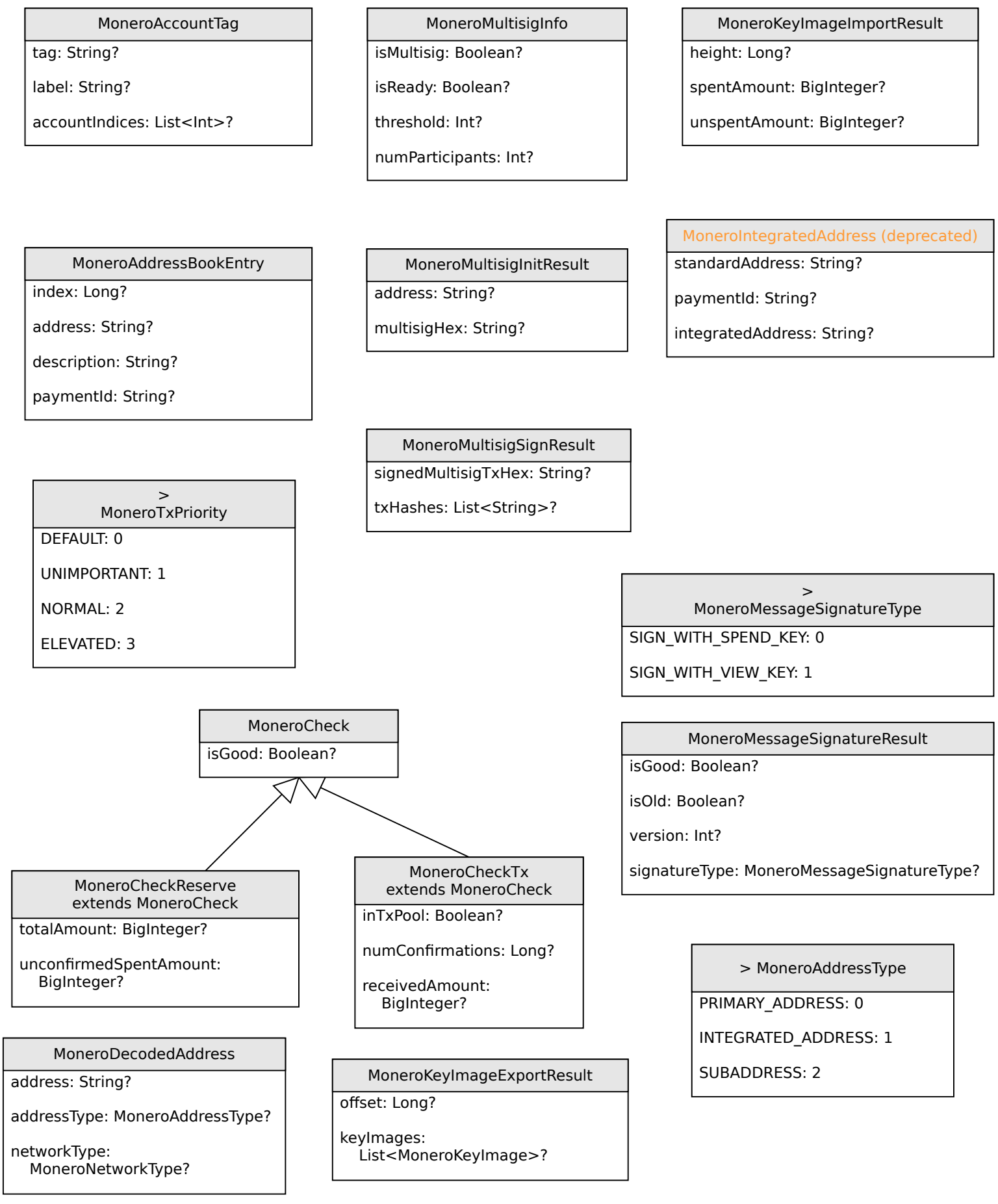
Monero Wallet Types 3/4



Monero Wallet Types 3/4 (continued)

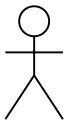


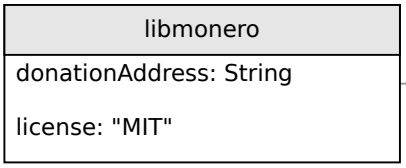
Monero Wallet Types 4/4





84DG2VwzuVjfmFS6vzfXmNdSGFSrsy922iuwntcCdLMcYe1tmUcsobz9ossuhCQT1YW3tTUFV2eUEAdgT9ANWaquPVeXZNE





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